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Studierichting: Informatica

Oefeningenreeks 2E nr. 4.7

$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\frac{ax + b}{ax + c} \right)^x \\ &= \lim_{x \rightarrow \infty} \left(\frac{ax + c - c + b}{ax + c} \right)^x \\ &= \lim_{x \rightarrow \infty} \left(1 + \frac{b - c}{ax + c} \right)^x \\ &= \lim_{x \rightarrow \infty} \left(\left(1 + \frac{b - c}{ax + c} \right)^{\frac{ax + c}{b - c}} \right)^{\frac{x(b - c)}{ax + c}} \\ &= \lim_{x \rightarrow \infty} e^{\frac{x(b - c)}{ax + c}} \\ &= e^{\frac{b - c}{a}} \text{ voor } a \neq 0 \end{aligned}$$

References